

Differentiation

Week 4A

Overview

Derivatives of other Functions

Key Ideas - Derivative of Logarithmic Functions

- *Derivative of Trigonometric Functions*
- *Derivative of Mixed of any Functions*

Key Idea – Derivative of Logarithmic Functions

The derivative of the function $y = \ln x$ is,

$$\frac{dy}{dx} = \frac{1}{x}$$

The chain rule allows us to differentiate,

$$y = f(\quad) = \ln(\quad), \text{ where } (\quad) \text{ is any differentiable function.}$$

Which gives us,

$$\text{If } y = \ln f(x) \text{ then } \frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

Remember:

“Differentiate the outside function first leaving the inside the same, then multiply by the derivative of what’s inside.”

Examples: Find the derivative of the following functions:

$$y = \log_e 3x$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{3}{3x} \\ &= \frac{1}{x}\end{aligned}$$

$$y = \log_e (3x - 5)$$

$$y' = \frac{3}{(3x - 5)}$$

$$y = 2 \ln(3x^2 - 5x)$$

$$\begin{aligned}y' &= 2 \times \frac{(3x - 5)}{(3x^2 - 5x)} \\ &= \frac{2(3x - 5)}{(3x^2 - 5x)}\end{aligned}$$

$$y = \ln 2x^3$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{6x^2}{2x^3} \\ &= \frac{3}{x}\end{aligned}$$

$$y = 2x^3 + \log_e (4x^2 - 1)$$

$$y' = 6x^2 + \frac{8x}{(4x^2 - 1)}$$

$$f(x) = \ln 4x^3 - 10$$

$$\begin{aligned}f'(x) &= \frac{12x^2}{4x^3} - 0 \\ &= \frac{3}{x}\end{aligned}$$

Exercise:

Find the derivative of the following:

(a) $y = \ln(2x)$
 $f(x) = 2x$
 $f'(x) = 2$

$$\frac{dy}{dx} = \frac{2}{2x} = \frac{1}{x}$$

$$\begin{aligned} y &= \ln f(x) \\ \frac{dy}{dx} &= \frac{f'(x)}{f(x)} \end{aligned}$$

(b) $y = 3\ln(x^2) - 4x^3$
 $f(x) = x^2$
 $f'(x) = 2x$

$$\begin{aligned} \frac{dy}{dx} &= 3 \times \frac{2x}{x^2} - 12x^2 \\ &= \frac{6}{x} - 12x^2 \end{aligned}$$

(c) $f(x) = \log_e(x^2 - 3x)$
 $f(x) = x^2 - 3x$
 $f'(x) = 2x - 3$

$$\frac{dy}{dx} = \frac{2x-3}{x^2-3x}$$

To differentiate logarithmic functions, it is often useful to firstly rewrite the function using the following log laws.

Laws of Logarithms:

- $\log_b xy = \log_b x + \log_b y$
- $\log_b \left(\frac{x}{y} \right) = \log_b x - \log_b y$
- $\log_b x^p = p \log_b x$

These laws can be used to simplify logarithms of any base, but the same base must be used throughout a particular calculation.

Examples: Find the derivative of the following functions:

(a) $y = \ln(x-6)^4$

$$\begin{aligned} y &= \ln(x-6)^4 \\ &= 4 \ln(x-6) \quad \text{using } \log_b x^p = p \log_b x \end{aligned}$$
$$\frac{dy}{dx} = \frac{4}{x-6}$$

$$(b)(i) \quad y = \ln \frac{1}{(3x+2)^4}$$

$$y = \ln \frac{1}{(3x+2)^4}$$

$$= \ln(3x+2)^{-4}$$

$$= -4 \ln(3x+2)$$

$$\frac{dy}{dx} = -\frac{12}{3x+2}$$

$$\text{using } x^a \times x^b = x^{a+b}$$

$$\log_b x^p = p \log_b x$$

$$(b)(ii) \quad y = \ln 3x\sqrt{x}$$

$$y = y = \ln(3x\sqrt{x}) = \ln\left(3x^{\frac{3}{2}}\right)$$

$$= \frac{3}{2} \ln(3x)$$

$$y' = \frac{3}{2} \times \frac{3}{3x} = \frac{3}{2x}$$

$$\text{using } x^a \times x^b = x^{a+b}$$

$$\log_b x^p = p \log_b x$$

$$(c) \quad y = \ln(1-3x)(1+2x)$$

$$\begin{aligned} y &= \ln(1-3x)(1+2x) \\ &= \ln(1-3x) + \ln(1+2x) \end{aligned}$$

$$y' = \frac{-3}{1-3x} + \frac{2}{1+2x}$$

$$\text{using } \log_b xy = \log_b x + \log_b y$$

$$(d) \quad y = \log_e \left(\frac{x^2}{x+4} \right)$$

$$\begin{aligned} y &= \log_e \left(\frac{x^2}{x+4} \right) \\ &= (\log_e x^2 - \log_e (x+4)) \end{aligned}$$

$$\text{using } \log_b \left(\frac{x}{y} \right) = \log_b x - \log_b y.$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{2x}{x^2} - \frac{1}{x+4} \\ &= \frac{2}{x} - \frac{1}{x+4} \\ &= \frac{2(x+4) - x}{x(x+4)} \\ &= \frac{x+8}{x(x+4)} \end{aligned}$$

Exercise:

Find the derivative of the following:

(a) $y = \ln(2x-3)^3$ use $\log_b x^p = p \log_b x$ for (a) & (b)

$$y' = 3 \ln(2x-3)$$
$$y' = 3 \times \frac{2}{2x-3} = \frac{6}{2x-3}$$

(b) $y = \ln(\sqrt{x-3}) - 4$

$$y = \ln(x-3)^{\frac{1}{2}} - 4$$
$$= \frac{1}{2} \ln(x-3) - 4$$

$$y' = \frac{1}{2} \times \frac{1}{x-3} = \frac{1}{2(x-3)}$$

(c) $y = \log_e \left(\frac{x-2}{x+1} \right)$ use $\log_b \left(\frac{x}{y} \right) = \log_b x - \log_b y$

$$y = \log_e(x-2) - \log_e(x+1)$$
$$y' = \frac{1}{x-2} - \frac{1}{x+1}$$

Exercise:

Find the derivative of the following:

(a) $y = \log_e(x-5)(x+3)$

use $\log_b(xy) = \log_b x + \log_b y$

$$y = \log_e(x-5) + \log_e(x+3)$$

$$y' = \frac{1}{x-5} + \frac{1}{x+3}$$

(b) $y = \ln \frac{1}{3x-1}$

$$= \ln(3x-1)^{-1}$$

use $\log_b x^p = p \log_b x$

$$y = -1 \ln(3x-1)$$

$$y' = -1 \times \frac{3}{3x-1} = \frac{-3}{3x-1}$$

Example using product rule: Differentiate $y = (x + 2)\log_e x$

$$y' = vu' + uv'$$

$$= \log_e x + (x + 2) \times \frac{1}{x}$$

$$= \log_e x + \frac{x + 2}{x}$$

$$u = x + 2$$

$$u' = 1$$

$$v = \log_e x$$

$$v' = \frac{1}{x}$$

Example using chain rule: Differentiate $y = \sqrt{\ln(2x - 1)}$

$$y = \sqrt{\ln(2x - 1)} = (\ln(2x - 1))^{\frac{1}{2}}$$

$$y' = \frac{1}{2}(\ln(2x - 1))^{-\frac{1}{2}} \times \left(\frac{2}{2x - 1} \right)$$

$$= \frac{1}{(2x - 1)\sqrt{\ln(2x - 1)}}$$

Example using quotient rule:

Differentiate $y = \frac{2x^2}{\log_e x}$

$$y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{4x \log_e x - 2x^2 \frac{1}{x}}{(\log_e x)^2}$$

$$= \frac{4x \log_e x - 2x}{(\log_e x)^2}$$

$$u = 2x^2$$

$$u' = 4x$$

$$v = \log_e x$$

$$v' = \frac{1}{x}$$

Exercise:

Find the gradient of the following functions:

(a) $y = x^3 \log_e 2x$ **product rule** $\frac{dy}{dx} = u'v + uv'$

$u = x^3$ $v = \log_e 2x$
 $u' = 3x^2$ $v' = \frac{2}{2x} = \frac{1}{x}$

$$\frac{dy}{dx} = 3x^2 \log_e 2x + x^3 \times \frac{1}{x}$$

$$= 3x^2 \log_e 2x + x^2$$

$$= x^2 (3 \log_e 2x + 1)$$

(b) $f(x) = \frac{e^2 \ln x - 1}{x - 1}$ **quotient**

$u = e^2 \ln x$ $v = x - 1$ $\frac{dy}{dx} = \frac{u'v - uv'}{v^2}$

$u' = \frac{e^2}{x}$ $v' = 1$

$$= \frac{\frac{e^2}{x} (x - 1) - e^2 \ln x \times 1}{(x - 1)^2}$$

$$= \frac{e^2 (x - 1 - \ln x)}{(x - 1)^2}$$

(c) $y = (\ln 2x^3 - 5x)^3$ **chain rule**

$$y' = 3(\ln 2x^3 - 5x)^2 \times \left(\frac{6x^2}{2x^3} - 5 \right)$$

$$y' = 3(\ln 2x^3 - 5x)^2 \left(\frac{3}{x} - 5 \right)$$

Exercise:

Find the gradient of the curve $y = \log_e(x^2 + 1)$ at the point where $x = 3$.

$$= \frac{dy}{dx}$$

$$y' = \frac{2x}{x^2 + 1}$$

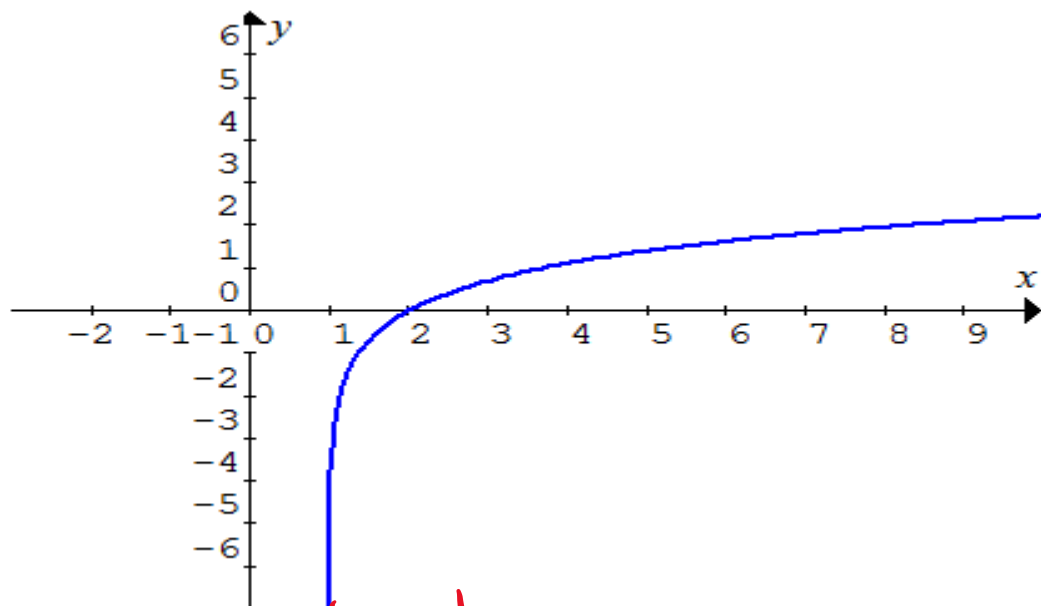
$$y'(3) = \frac{2 \times 3}{3^2 + 1}$$

$$= \frac{6}{10}$$

$$= \frac{3}{5}$$

Exercise:

The graph of $y = \ln(x-1)$ is drawn below. Find the equation of the tangent at the point $x = 2$.



$$\bullet m = y' = \frac{1}{x-1}$$

$$\text{at } x=2: m = y'(2) = \frac{1}{2-1} = 1$$

$$\bullet y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x - 2)$$

$$y = x - 2$$

need to use
 $y - y_1 = m(x - x_1)$

- find y-coord
sub $x=2$ into
 $y = \ln(x-1)$
 $\therefore y(2) = \ln(2-1)$
 $= \ln 1$
 $= 0$
 $\therefore (2, 0)$

Key Idea – Derivative of Trigonometric Functions

The derivative of the trigonometric functions are,

$$\text{If } y = \sin x \text{ then } \frac{dy}{dx} = \cos x$$

$$\text{If } y = \cos x \text{ then } \frac{dy}{dx} = -\sin x$$

$$\text{If } y = \tan x, \frac{dy}{dx} = \sec^2 x$$

The chain rule allows us to differentiate,

$$y = f(\quad) = \sin(\quad), \quad y = f(\quad) = \cos(\quad) \text{ or } y = f(\quad) = \tan(\quad)$$

where $(\quad)$ is any differentiable function.

Which gives us,

$$\text{If } y = \sin f(x) \text{ then } \frac{dy}{dx} = f'(x) \cos f(x)$$

$$\text{If } y = \cos f(x) \text{ then } \frac{dy}{dx} = -f'(x) \sin f(x)$$

$$\text{If } y = \tan f(x) \text{ then } \frac{dy}{dx} = f'(x) \sec^2 f(x)$$

Examples: Differentiate the following:

$$y = \sin 2x$$

$$\frac{dy}{dx} = 2 \cos(2x)$$

$$y = \sin(x^2) + \cos(3x + 3)$$

$$\frac{dy}{dx} = 2x \cos(x^2) - 3 \sin(3x + 3)$$

$$y = 3x + 2 \tan \frac{x}{2}$$

$$\begin{aligned} \frac{dy}{dx} &= 3 + 2 \times \frac{1}{2} \sec^2 \frac{x}{2} \\ &= 3 + \sec^2 \frac{x}{2} \end{aligned}$$

$$y = 2x - \cos(4x + 3)$$

$$\frac{dy}{dx} = 2 + 4 \sin(4x + 3)$$

$$f(x) = \cos(2x + \pi)$$

$$f'(x) = -2 \sin(2x + \pi)$$

$$y = \tan(3x + 1)$$

$$\frac{dy}{dx} = 3 \sec^2(3x + 1)$$

Exercise:

Write the down $\frac{dy}{dx}$ for each of the following:

(a) $y = \sin 2x$

$$f(x) = 2x$$

$$f'(x) = 2$$

$$\frac{dy}{dx} = 2 \cos 2x$$

(b) $y = \sin x^2$

$$f(x) = x^2$$

$$f'(x) = 2x$$

$$\frac{dy}{dx} = 2x \cos x^2$$

(c) $y = 2 \cos(x^2 + x)$

$$f(x) = x^2 + x$$

$$f'(x) = 2x + 1$$

$$\frac{dy}{dx} = -(2x + 1) \sin(x^2 + x)$$

Exercise:

Write the down $\frac{dy}{dx}$ for each of the following:

(a) $y = 3 \tan x - \sin 4x^3$

$$\begin{aligned} f(x) &= x & f(x) &= 4x^3 & \frac{dy}{dx} &= 3 \sec^2 x - 12x^2 \cos(4x^3) \\ f'(x) &= 1 & f'(x) &= 12x^2 \end{aligned}$$

(b) $y = \tan \frac{x}{4} - 3x^2$

$$\begin{aligned} f(x) &= \frac{x}{4} = \frac{1}{4}x & \frac{dy}{dx} &= \frac{1}{4} \sec^2\left(\frac{x}{4}\right) - 6x \\ f'(x) &= \frac{1}{4} \end{aligned}$$

Examples using chain rule:

$$y = (\cos x)^2$$

$$\begin{aligned} y' &= 2(\cos x)(-\sin x) \\ &= -2\cos x \sin x \end{aligned}$$

$$f(x) = (\sin 5x)^3$$

$$\begin{aligned} f'(x) &= 3(\sin 5x)^2 (\cos 5x \times 5) \\ &= 15\cos 5x (\sin 5x)^2 \end{aligned}$$

Example using product rule:

$$f(x) = \sin 2x \cos 2x$$

$$f'(x) = u'v + uv'$$

$$= -2\cos 2x \times \cos 2x + \sin 2x \times (-2)\sin 2x$$

$$= -2\cos^2 2x - 2\sin^2 2x$$

$$= -2(\cos^2 2x + \sin^2 2x)$$

$$= -2 \times 1$$

$$= -2$$

$$u = \sin 2x$$

$$u' = -2\cos x$$

$$v = \cos 2x$$

$$v' = -2\sin 2x$$

Example using quotient rule:

$$y = \frac{\sin x}{x^2}$$

$$u = \sin x$$
$$u' = -\cos x$$

$$v = x^2$$
$$v' = 2x$$

$$y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{x^2 \times (-\cos x) - \sin x \times 2x}{(x^2)^2}$$

$$= \frac{-x^2 \cos x - 2x \sin x}{x^4}$$

$$= \frac{-x(x \cos x + 2 \sin x)}{x^4}$$

$$= \frac{-(x \cos x + 2 \sin x)}{x^3}$$

Exercise:

Differentiate the following:

product rule: $y = u \times v$
 $y' = u'v + uv'$

(a) $f(x) = \sin x(1 - \cos x)$

$u = \sin x$ $u' = \cos x$ $v = 1 - \cos x$ $v' = \sin x$
 $y' = \cos x(1 - \cos x) + \sin x \times (\sin x)$
 $y' = \cos x - \cos^2 x + \sin^2 x$

(b) $y = \sqrt{\tan(8x-1)}$

$y = (\tan(8x-1))^{\frac{1}{2}}$
 $y' = \frac{1}{2}(\tan(8x-1))^{-\frac{1}{2}} \times (8 \sec^2(8x-1))$
 $= \frac{4 \sec^2(8x-1)}{\sqrt{\tan(8x-1)}}$

(c) $y = \frac{x}{\sin 2x}$

quotient rule
 $y = \frac{u}{v}$ $y' = \frac{u'v - uv'}{v^2}$

$u = x$ $v = \sin 2x$

$u' = 1$ $v' = 2 \cos 2x$

$y' = \frac{1 \times \sin 2x - x \times 2 \cos 2x}{(\sin 2x)^2}$
 $= \frac{\sin 2x - 2x \cos 2x}{\sin^2 2x}$

Discussion Question:

Is $\frac{d}{dx} \cos^3 x = \frac{d}{dx} \cos x^3$, justify your answer?

$$\begin{aligned}\frac{d}{dx}(\cos^3 x) &= \frac{d}{dx}((\cos x)^3) \\ &= 3(\cos x)^2 \times -\sin x \\ &= -3\sin x (\cos x)^2 \\ &= -3\sin x \cos^2 x\end{aligned}$$

$$\frac{d}{dx}(\cos \overset{\uparrow}{x^3}) = -3x^2 \sin x^3$$

$$\begin{aligned}f(x) &= x^3 \\ f'(x) &= 3x^2\end{aligned}$$

$$\text{Note: } \cos^3 x = [\cos x]^3 \\ [\cos x \times \cos x \times \cos x]$$

$$\begin{aligned}\text{Note: } \cos x^3 &= \cos(x \times x \times x) \\ \cos x^3 &\neq \cos^3 x\end{aligned}$$

$$\therefore \text{ No } \frac{d}{dx}(\cos^3 x) \neq \frac{d}{dx}(\cos x^3)$$

Exercise:

Find the equation of the tangent to the curve $\tan x$ at the point where $x = \pi$.

we need to use $y - y_1 = m(x - x_1)$

- $x_1 = \pi$

- y_1 : sub $x_1 = \pi$ into

$$y = \tan x$$
$$y = \tan \pi$$
$$= 0$$

- $m = y' = \sec^2 x$

at $x = \pi$

$$m = y'(\pi) = \sec^2(\pi)$$

$$= \frac{1}{\cos^2(\pi)}$$

$$= \frac{1}{(\cos \pi)^2}$$

$$= \frac{1}{(-1)^2}$$

$$= 1$$

$$\therefore y - y_1 = m(x - x_1)$$

$$y - 0 = 1(x - \pi)$$

$$\underline{\underline{y = x - \pi}}$$

Key Idea – Derivative of any Functions

The following exercises are a combination of product rule, quotient rule and chain rule using all the different functions introduced in this course.

Product Rule: $\frac{d}{dx}(uv) = u'v + uv'$

Quotient Rule: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$

Chain Rule: $\frac{d}{dx}[f(g(x))] = f'(g(x)) \times g'(x)$

Examples: Find the derivative of the following:

$$\begin{aligned}\frac{d}{dx}e^{\sin 2x} &= 2\cos 2x \times e^{\sin 2x} \\ &= 2e^{\sin 2x} \cos 2x\end{aligned}$$

$$\begin{aligned}\frac{d}{dx}\ln(\sin 2x) &= \frac{2\cos 2x}{\sin 2x} \\ &= 2\cot 2x\end{aligned}$$

Exercise:

Find the derivative of the following:

(a) $y = e^{-x} \cos 3x$ product rule $y' = u'v + uv'$
 $u = e^{-x}$ $u' = -e^{-x}$ | $v = \cos 3x$ $v' = -3\sin 3x$

$$y' = -e^{-x} \times \cos 3x + e^{-x} \times (-3\sin 3x)$$
$$= -e^{-x} (\cos 3x + 3\sin 3x)$$

(b) $y = \frac{2 \tan 3x}{x^2}$ quotient rule $y' = \frac{u'v - uv'}{v^2}$

$$u = 2 \tan 3x \quad u' = 6 \sec^2 3x \quad | \quad v = x^2 \quad v' = 2x$$

$$y' = \frac{6 \sec^2 3x \times x^2 - 2 \tan 3x \times 2x}{(x^2)^2}$$

$$= \frac{6x^2 \sec^2 3x - 4x \tan 3x}{x^4}$$

$$= \frac{6x \sec^2 3x - 4 \tan 3x}{x^3}$$

(c) $y = 3\ln(\sin(2x^3))$ chain

$$y' = 3 \times \frac{6x^2 \cos(2x^3)}{\sin(2x^3)}$$

$$= \frac{18x^2 \cos(2x^3)}{\sin(2x^3)} = 18x^2 \cot(2x^3)$$

(d) $f(x) = (2x^2 - 3e^{2x} + 4\tan x)^3$ chain

$$\begin{aligned} f'(x) &= 3(2x^2 - 3e^{2x} + 4\tan x)^2 \times \text{derivative of } (2x^2 - 3e^{2x} + 4\tan x) \\ &= 3(2x^2 - 3e^{2x} + 4\tan x)^2 \times (4x - 6e^{2x} + 4\sec^2 x) \\ &= 6(2x - 3e^{2x} + 2\sec^2 x)(2x^2 - 3e^{2x} + 4\tan x)^2 \end{aligned}$$